The Failure Point of Systems of Linearized Differential Equations

How far from equilibrium does the linearization of a two-variable energy-balance climate model remain a valid predictor of the system's behaviour?

International Baccalaureate Investigation in Mathematics

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3800 Words

Contents

1 Introduction **1**

1.1 Game explained 1

1.1.1 Setup 1

1.1.2 Rules 2

1.2 Definitions 2

1.3 Aim and Methodology 3

1.4 Assumptions 3

2 Exploration **4**

2.1 Initial Trials 4

2.2 Data Collection 5

2.3 Graphing 5

1 Introduction

Some patterns are hard to turn a blind eye on, especially when they might sway the outcome one way or another. While playing my favourite card game ”Speed”, I noticed that the more rounds of a game we played, the more ordered the cards became, which made me wonder if this was in fact true, or if I had jumped to logical conclusions. Playing cards is one of my favourite ways to connect with people, and so I wanted to explore the maths behind this game.

1.1 Game explained

1.1.1 Setup

In the game of ”Speed”, the deck is split evenly in half, and each player sets up their half of the deck, respectively, in a layered formation, as shown below:

Figure 1: first layer - one faced up, four faced down

Figure 2: second layer - one faced up, three faced down

Figure 3: third layer - one faced up, two faced down

Figure 4: fourth layer - one faced up, one faced down

Figure 5: fifth layer - one faced up

The remainder of each half deck is placed in the centre, and now act as draw piles, that are played alongside onto the respective playing pile:

Figure 6: final setup of Speed

1.1.2 Rules

The aim of the game is to finish the layered cards in front of you. A card can be played if it is either one above or one below (not the same) the card shown on the playing pile, and suits don't matter.

Figure 7: a good sample (more ordered) of starting cards

Figure 8: a bad sample (less ordered) of starting cards

Thus, the more similar in number your cards are, the more of them you can play. Each turn, both players draw a card, and play out their own cards onto either pile until neither can play anymore, then another card is drawn and the turns repeat until one player depletes their cards. At this point, either player can select the smallest pile in front of them, slap it, and claim it as part of their cards, while the loser is condemned to the larger pile. These newly claimed piles are added on top of each respective draw pile, without being shuffled, and the players set up another round within the larger game.

1.2 Definitions

- Number of rounds played (a): the rounds played within one game, with no shuffling between each one
- Cards flipped from the draw pile (b): The number of cards flipped, simultaneously, by both players
- Length of longest chain (c): the number of cards that are either one up or one down from the previous card creating a chain, applicable for this specific game

1.3 Aim and Methodology

This investigation aims to explore the relationship between the number of rounds played and the orderedness of the cards, by enacting the following steps: 1. Collecting valid and reliable data 2. Graphing the data, and assessing which regression line fits the data points best. Understanding and outlining the reasons for this pattern in the data. 3.

1.4 Assumptions

The findings made in this investigation are contextualised by 1) control variables such as all games (data) are played by the same two players and by 2) ideal conditions in which both players maintain their usual skill level throughout all games so as not to disrupt the data.

2 Exploration

2.1 Initial Trials

With the aim of understanding how the number of rounds affects the orderedness of the cards in play, I first collected data on the relationship between the following variables:

Axis	Description	Variable
x	The number of rounds played	Independent
y	The cards flipped from the draw pile	Dependent

Table 1: PLACEHOLDER

The cards flipped from the draw pile is a key indicator that isolates the orderedness of only the cards in play rather than assessing the orderedness of the whole deck, reducing the scope and therefore increasing the validity. However, after collecting this data and graphing it on Desmos, I discovered that the relationship gave a random result that didn't indicate any kind of trend, like the one I had hypothesised.

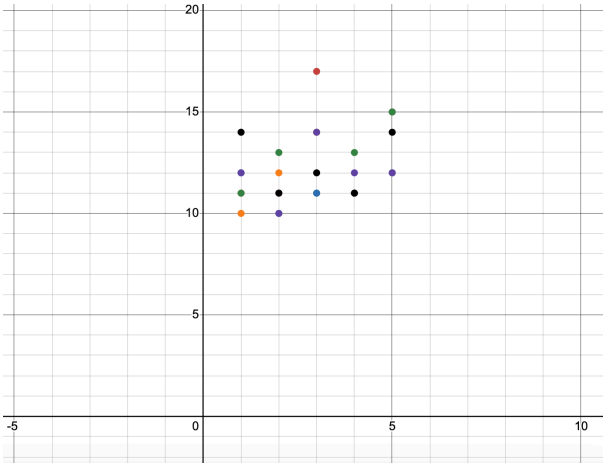


Figure 1: First Fail

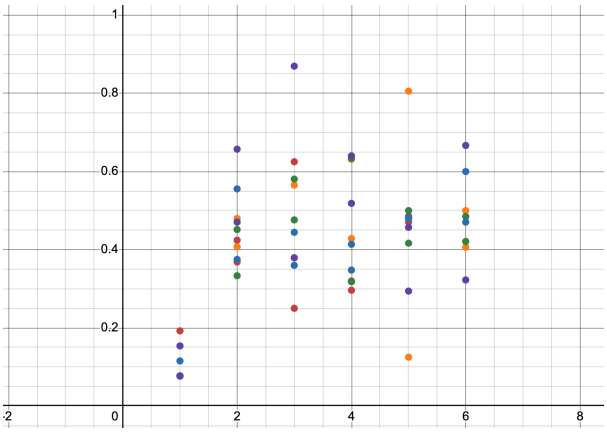


Figure 2: Second Fail

Following this setback, I tried a different approach that still assessed the number of

rounds as the independent variable, however changed the dependent variable to include the whole deck of cards, rather than just the few cards in play as assesssing the orderedness of the few cards faced up in front of each player could be useful for individual rounds, but the goal of assessing a whole game's progress inevitably requires analysis of the whole deck. Additionally, I changed the actual determining factor to be the longest run that exists in each player's hand, rather than the number of cards drawn, to more accurately represent the orderedness of cards as it allows for cards in order, both ascending and descending, to be recorded, rather than the indicator of cards drawn which can get skewed by the cards that players have available.

In this new exploration, I used the following factors:

Independent variable: number of the round (n)
Dependent variable: length of the longest chain (c)

2.2 Data Collection

I played 10 games of Speed, limiting each game to 5 rounds. I took count of the c value before each round, and once after the final, fifth round, producing 6 data points for each game.

Figure 9: Sample of the data found (see Appendix for full data set)

2.3 Graphing

After collecting adequate data on these variables, I found the average ratio (r), for each round from 0-5, for each player individually (r1 and r2), and then found the average between those, which presented the data set:

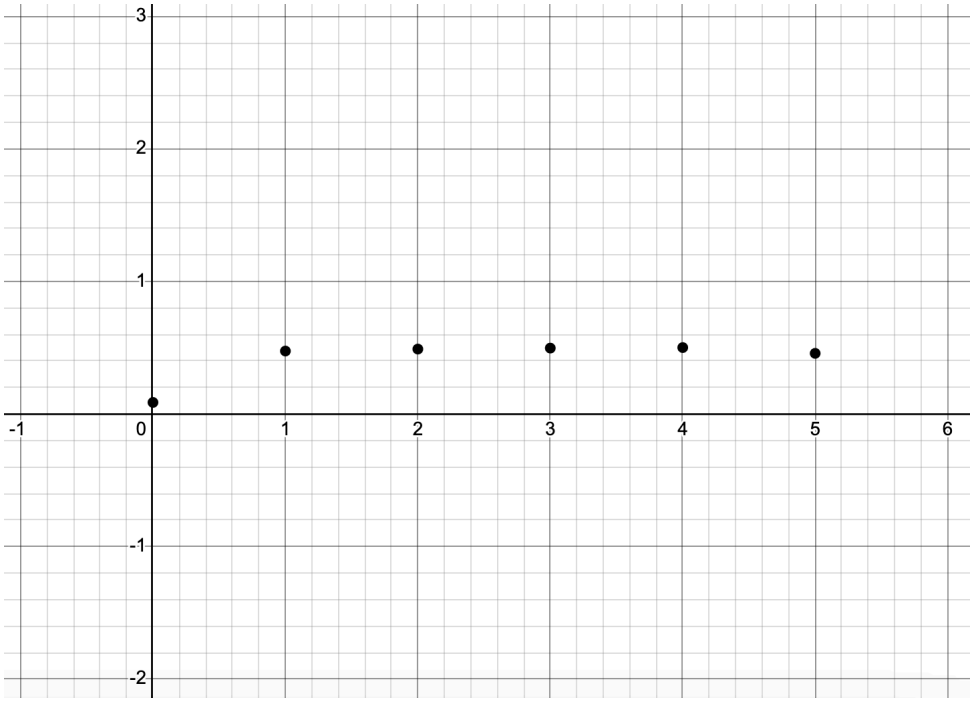


Figure 3: Graph of averaged values

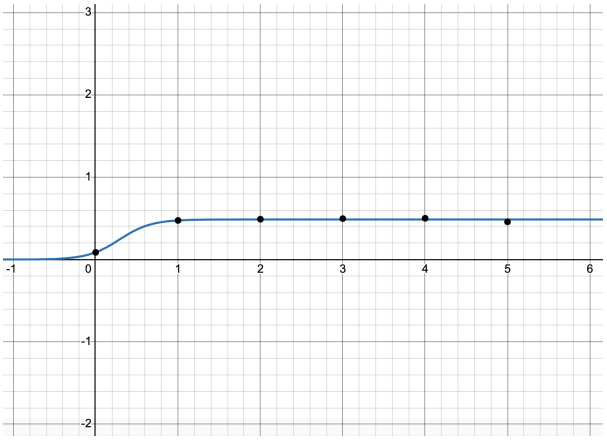


Figure 4: Graph with logistic regression line

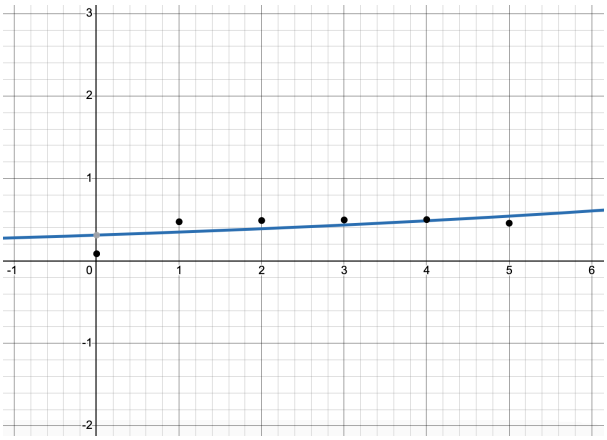


Figure 5: Graph with exponential regression line

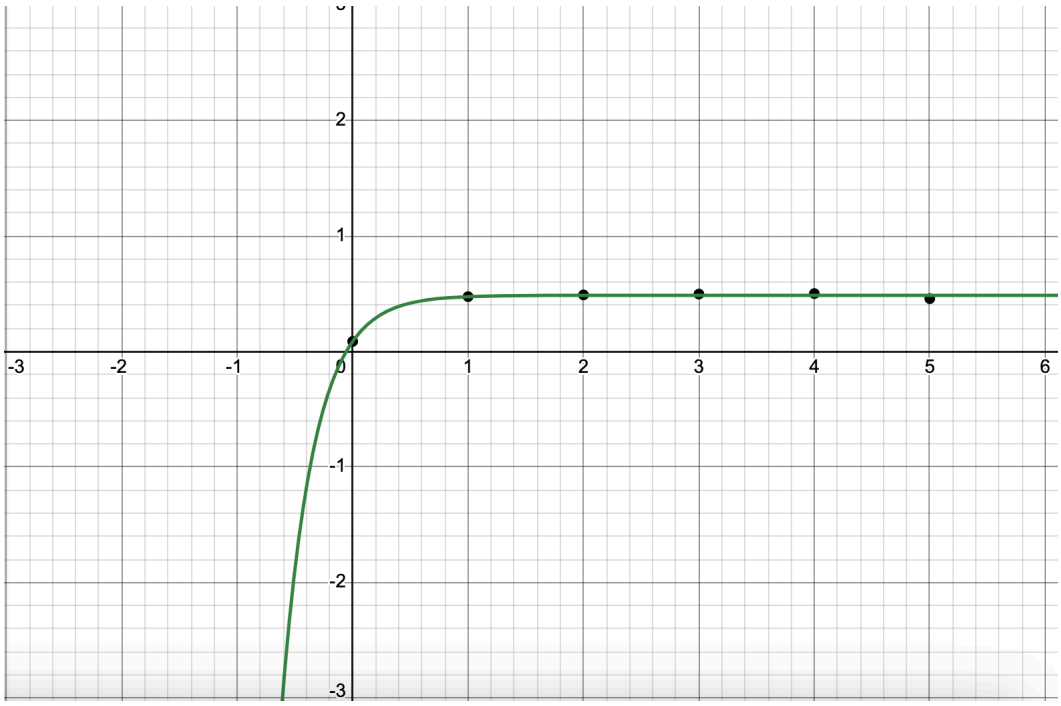


Figure 6: Graph with line of best fit

I tried various regression lines on this data (Fig. 12-14), however evidently the line of best fit was the exponential graph (Fig. 12) with the following equation:

This graph illustrates the extreme incline and then rounding and ultimately flat-lining of the data best, especially in comparison to the others. This curve also begins by going directly up, rather than gradually ascending, such as in Fig. 13. From this graph, we learn that the relationship between the number of rounds played and the number of cards in the longest chain is weak in the first round, however thereafter increases sharply and settles around 0.45-0.5 without dramatically increasing or decreasing, indicating n asymptotic nature of the orderedness of cards in Speed rounds. Through a real-world lens, this ultimately indicates that while the more rounds played does result in more ordered cards, this does not perpetuate the cycle entirely until the entire deck is ordered, but rather cuts off once the deck reaches a certain level of orderedness.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{1}$$

in equation 1 the denominator is $2a$.

(Be explicit about mistakes, new decisions, disregarding a certain approach etc.)

abc bc can be grouped d can be asymptote